

FRC 2025 REEFSCAPE — Top Robot Design Deep Dive

A study reference for effective REEFSCAPE design & implementation, drawn from the Spectrum 3847 CAD Collection, team CAD/code releases (Onshape / GrabCAD), and each team's Chief Delphi reveal / CAD-release thread.

The REEFSCAPE design landscape

The game

REEFSCAPE is a coral-and-algae game: robots place CORAL (PVC tubes) on the REEF's four branch levels (L1-L4) and score ALGAE (foam balls) into the PROCESSOR or BARGE NET, then climb a CAGE in the endgame. Most top designs centered on a fast coral cycle (station or ground intake -> elevator/arm -> reef); many top teams ran coral-only and treated algae and deep climb as optional, lower-value choices.

How to use this document

These are the most thoroughly-documented REEFSCAPE robots from the CAD-release set, each with real subsystem detail mined from the team's Chief Delphi reveal/release thread. For every team we capture archetype, the four subsystems (scoring / intake / elevator-lift / drivetrain), vision, public CAD availability, and 'lessons for students'. ★ marks the curated study picks with the richest reproducible documentation. EPA / rank are pending a Statbotics API outage and will be backfilled when it recovers.

Profiled robots at a glance

Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★ 1690 Orbit	WHISPER	—	—	Closed-belt differential elevator; fast coral cycle	Onshape CAD
★ 1114 Simbotics	Simbot Suzuki	—	—	Coral-only, station-intake elevator (deliberately minimal scop	Onshape CAD
★ 695 Bison Robotics	Goldfish	—	—	Fully-autonomous coral scorer; repulsion-field auto-align	Onshape CAD
★ 341 Miss Daisy	(2025)	—	—	Floor-intake continuous elevator; lantern-gear shoulder + wris	CD thread
★ 1086 Blue Cheese	Stingray	—	—	Full-capability: 4-level coral, algae net+processor, deep clim	CD thread
★ 1153 Timberwolves	Lagotto	—	—	1-stage elevator (L1-L3) + algae; ground algae; sub-15 s deep	CD thread
★ 3005 RoboChargers	Relay	—	—	3-stage elevator + 'laterator' end-effector (coral ejector + a	CD thread
★ 2930 Sonic Squirrels	(2025)	—	—	Coral robot; deep Choreo auto library + PhotonVision	CD thread
★ 1732 Hilltopper Robotics	(2025)	—	—	Coral arm/claw; 4x-L4 autonomous; algae handoff to processor	CD thread
★ 2910 Jack in the Bot	Spectre	—	—	Very compact self-aligning ground coral intake; fast cycle	CD thread
★ 8159 Golden Horn	Serpentheim	—	—	Coral robot, dual Limelight; 2 regional wins + first Impact Aw	CD thread
★ 190 Gompei and the HERD	(2025 V2)	—	—	6-motor elevator; star-wheel outtake flips coral up onto L4	CD thread
★ 1156 Under Control	(2025)	—	—	Swerve coral robot; permanent orientation-PID heading hold; te	CD thread
★ 118 Robonauts	Firefly	—	110-12-0	Coral + algae robot; 118 also author the 2025 Everybot	GrabCAD

	Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★	422 The Mech Tech Dragons	Wisp	—	62-6-0	Coral robot: ground coral intake -> elevator -> end effector,	Onshape CAD
★	5940 BREAD	Taiyaki	—	47-4-0	Coral + algae robot: full-width intake -> star-wheel indexer -	Onshape CAD
★	7407 Wired Boars	Griptide	—	55-13-0	Elevator-based coral robot that also REMOVES ALGAE; Open Allia	Onshape CAD

★ = recommended study pick (downloadable CAD or a full technical binder + a clear design lesson). EPA shown is Local EPA (approx), computed from The Blue Alliance match data.

Team-by-team profiles

★ Study pick

1690 Orbit WHISPER
ISR

EPA pending

Closed-belt differential elevator; fast coral cycle

SCORING (CORAL / ALGAE) Coral end-effector fed off the differential elevator for L1-L4 placement; among the fastest cycles in the world that season.	INTAKE Coral handling tuned for a rapid, repeatable cycle (see CAD).
ELEVATOR / LIFT Closed-belt DIFFERENTIAL elevator — two motors drive a closed belt loop so the carriage height and a second axis are co-controlled; credited to a community design Orbit adapted.	DRIVETRAIN Swerve.
ENDGAME Cage climb (deep).	AUTONOMOUS Fast multi-coral auto enabled by the quick cycle (count unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A closed-belt differential elevator couples two motions through one belt loop — compact and stiff.
- Strong results come from cycle-time focus, not feature count.

Onshape CAD: <https://cad.onshape.com/documents/76609fe05a6594c5f9c4062a/w/437a97728f1629348e9dd7cf/e/465d3a35c190dab8355f1e5c> · CD reveal — WHISPER · CD CAD release

★ Study pick

1114 Simbotics Simbot Suzuki
CAN

EPA pending

Coral-only, station-intake elevator (deliberately minimal scope)

SCORING (CORAL / ALGAE) Coral end-effector placing on the reef; team explicitly chose NOT to do ground algae or floor coral.	INTAKE Human-player STATION intake for coral (no floor intake by design).
ELEVATOR / LIFT Elevator superstructure; manipulator position recovery was a known failure mode they discuss in the release.	DRIVETRAIN Swerve.
ENDGAME By design, NO deep climb — scope was dropped to perfect the coral cycle (park only).	AUTONOMOUS Coral preload + station-fed scoring; deliberately simple.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- Scope discipline: dropping deep climb, ground algae and floor coral let them perfect one cycle.
- Document your failure modes (lost manipulator position) — it teaches more than a highlight reel.

Onshape CAD: <https://cad.onshape.com/documents/8782ccd3db870fc41090b5e6/w/88d1084497e586d2c58a4051/e/f1dc3344d36d78907a60d4d9> · CD CAD release

Fully-autonomous coral scorer; repulsion-field auto-align

SCORING (CORAL / ALGAE) Coral end-effector with fully autonomous scoring across reef levels.	INTAKE Autonomous feeder-station pickup (no driver input needed for the cycle).
ELEVATOR / LIFT Elevator superstructure; all mechanism control loops run on-board the motor controllers.	DRIVETRAIN Full Kraken-X60 swerve drivebase.
ENDGAME Cage climb.	AUTONOMOUS Fully autonomous coral scoring — MegaTag2 + 200 Hz odometry, repulsion-field auto-align, no driver input.
VISION Dual Limelight 4 running MegaTag2; 200 Hz odometry fused with a NavX2 for robust localization.	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- MegaTag2 + high-rate odometry (200 Hz) makes full-auto scoring practical.
- A repulsion (potential) field is a clean way to auto-align while avoiding field collisions.
- Running control loops on the motor controllers offloads the roboRIO.

Onshape CAD: <https://cad.onshape.com/documents/952ba22fa909a3d0882388da/w/94a37671574438408b83a5fc/e/97e98c3970941d33c368b387> · [CD CAD & code release](#)

Floor-intake continuous elevator; lantern-gear shoulder + wrist; deep climb

SCORING (CORAL / ALGAE) Coral end-effector on a lantern-gear-driven SHOULDER with a 180-degree WRIST for multi-level reef placement.	INTAKE Coral FLOOR intake.
ELEVATOR / LIFT Continuous (single-stage continuous) elevator; a software 'superstructure' supervises arm/elevator states to prevent self-collision.	DRIVETRAIN Swerve.
ENDGAME Deep cage climb.	AUTONOMOUS Floor-intake multi-level coral auto; the superstructure state machine prevents self-collision.
VISION Unconfirmed (check CAD/code)	PUBLIC CAD / DESIGN CD CAD/Code/Docs release

LESSONS FOR STUDENTS

- A state-supervising 'superstructure' layer prevents the arm from colliding with the robot — model your mechanism states explicitly.
- Floor intake + continuous elevator + 180-degree wrist is a flexible multi-level coral solution.

CD thread: <https://www.chiefdelphi.com/t/500739> · [CD CAD, code & documentation release](#) · [Onshape CAD](#)

★ Study pick

1086 Blue Cheese Stingray

USA-VA

EPA pending

Full-capability: 4-level coral, algae net+processor, deep climb

SCORING (CORAL / ALGAE) Coral on all four reef levels; algae removed from the reef and scored in both the net and the processor.	INTAKE Coral from the human station; algae off the reef.
ELEVATOR / LIFT Elevator superstructure carrying the coral/algae effector.	DRIVETRAIN Swerve.
ENDGAME Deep cage climb (part of the do-everything scope).	AUTONOMOUS Attempts a broad coral/algae auto; execution lagged the ambition (see the ambition-vs-execution note).
VISION Unconfirmed (check CAD/code)	PUBLIC CAD / DESIGN CD CAD & code release

LESSONS FOR STUDENTS

- Study this as an ambition-vs-execution case, not a design exemplar: a genuine do-everything robot (4-level coral + algae net/processor + deep climb) is buildable, but here the scope outran on-field results (EPA rank ~657/3690).
- The lesson is scope discipline — weigh how much capability a team can realistically build, debug, and drive reliably before committing to do-everything.
- Releasing CAD + code together still makes the whole ambitious system reproducible to learn from.

CD thread: <https://www.chiefdelphi.com/t/506376> · CD CAD & code release · Onshape CAD

★ Study pick

1153 Timberwolves Lagotto

EPA pending

1-stage elevator (L1-L3) + algae; ground algae; sub-15 s deep climb

SCORING (CORAL / ALGAE) Coral on L1-L3; algae scored into the processor and barge.	INTAKE Ground ALGAE intake (plus coral handling).
ELEVATOR / LIFT Single-stage elevator — simpler than multi-stage rivals, traded top-level reach for reliability.	DRIVETRAIN Swerve.
ENDGAME Sub-15-second deep cage climb — used as endgame insurance.	AUTONOMOUS L1-L3 coral + ground-algae auto (count unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN CD CAD release

LESSONS FOR STUDENTS

- A 1-stage elevator can win districts if the rest of the cycle is fast and reliable.
- A sub-15-second deep climb is a strong endgame insurance policy.

CD thread: <https://www.chiefdelphi.com/t/502263> · CD CAD release · Onshape CAD

★ Study pick

3005 RoboChargers Relay

USA-TX

EPA pending

3-stage elevator + 'laterator' end-effector (coral ejector + algae gripper)

SCORING (CORAL / ALGAE) A 'laterator' carries both a coral ejector and an algae gripper; coral is routed THROUGH the elevator to the effector.	INTAKE Funnel-assisted coral handling; algae gripper for reef/processor.
ELEVATOR / LIFT 3-stage elevator carrying the laterator.	DRIVETRAIN Swerve.
ENDGAME Cage climb.	AUTONOMOUS Coral auto routed through the elevator to the laterator (count unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN CD reveal — Relay

LESSONS FOR STUDENTS

- Routing the game piece through the elevator saves a separate hand-off path.
- A 'laterator' (lateral translator) end-effector neatly combines two scoring jobs.

CD thread: <https://www.chiefdelphi.com/t/493529> · CD reveal — Relay · Onshape CAD

★ Study pick

2930 Sonic Squirrels (2025)

USA-WA

EPA pending

Coral robot; deep Choreo auto library + PhotonVision

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	INTAKE Coral intake feeding an indexer (binder: Intake / Indexer sections).
ELEVATOR / LIFT Elevator superstructure.	DRIVETRAIN Swerve.
ENDGAME Cage climb.	AUTONOMOUS Large library of short Choreo paths composed into flexible coral autos; PhotonVision-localized.
VISION PhotonVision on two Orange Pi 5 coprocessors; 100+ short Choreo paths stitched into driver-selectable autos.	PUBLIC CAD / DESIGN CD CAD & code release

LESSONS FOR STUDENTS

- Build autos from many SHORT Choreo paths so the drive team can compose routines on the fly.
- Dual coprocessors give you redundant / wider-FOV PhotonVision coverage.

CD thread: <https://www.chiefdelphi.com/t/505039> · CD CAD & code release · Onshape CAD

★ Study pick

1732 Hilltopper Robotics (2025)

USA-KY

EPA pending

Coral arm/claw; 4x-L4 autonomous; algae handoff to processor

SCORING (CORAL / ALGAE) Coral claw placing on the reef (up to four L4 in auto); algae handed down into the processor.	INTAKE Coral intake feeding the claw; algae path with added guarding to keep coral out.
ELEVATOR / LIFT Arm/lift superstructure.	DRIVETRAIN Swerve.
ENDGAME Cage climb; algae handed to the processor.	AUTONOMOUS 4x L4 coral autonomous; QuestNav (VR inside-out) localization.
VISION QuestNav (Meta Quest headset) used for localization in the 4x-L4 auto.	PUBLIC CAD / DESIGN CD reveal (4x L4 QuestNav auto)

LESSONS FOR STUDENTS

- QuestNav (a VR headset's inside-out tracking) is a low-cost, high-rate localization source for autos.
- Design the algae path so it can't accidentally ingest coral.

CD thread: <https://www.chiefdelphi.com/t/492785> · CD reveal — 4x L4 QuestNav auto · Onshape CAD

★ Study pick

2910 Jack in the Bot Spectre

USA-WA

EPA pending

Very compact self-aligning ground coral intake; fast cycle

SCORING (CORAL / ALGAE) Coral end-effector for reef placement; high throughput (paired with 9442 at events).	INTAKE One of the smallest self-aligning GROUND coral intakes in the game — centers the coral as it ingests.
ELEVATOR / LIFT Elevator superstructure.	DRIVETRAIN Swerve.
ENDGAME Cage climb.	AUTONOMOUS Fast coral auto from the self-centering ground intake (count unconfirmed).
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN CD CAD release

LESSONS FOR STUDENTS

- Passive geometry can self-center a game piece, shrinking the intake envelope.
- A small, reliable ground intake enables a faster, more flexible cycle.

CD thread: <https://www.chiefdelphi.com/t/500310> · CD reveal — Spectre · CD CAD release · Onshape CAD

★ Study pick

8159 Golden Horn Serpentheim

TUR

EPA pending

Coral robot, dual Limelight; 2 regional wins + first Impact Award

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	INTAKE Coral intake (see CAD).
ELEVATOR / LIFT Elevator superstructure.	DRIVETRAIN Swerve.
ENDGAME Cage climb.	AUTONOMOUS Strategy-driven coral auto (dual Limelight); see released strategy docs.
VISION Two Limelight cameras (discussed in the release Q&A).	PUBLIC CAD / DESIGN CD CAD, code & strategy release

LESSONS FOR STUDENTS

- Pairing CAD + code + STRATEGY documents is rare and valuable — study how strategy drove their design.
- Strong engineering and outreach can come together (2 regionals + first Impact Award).

CD thread: <https://www.chiefdelphi.com/t/509509> · CD CAD, code & strategy release · Onshape CAD

★ Study pick

190 Gompei and the HERD (2025 V2)

USA-MA

EPA pending

6-motor elevator; star-wheel outtake flips coral up onto L4

SCORING (CORAL / ALGAE) Outtake with STAR WHEELS that flip the coral upward as it leaves, seating it on L4; a ramp straightens/holds the coral instead of a powered belt.	INTAKE Coral intake feeding the ramp.
ELEVATOR / LIFT 6-motor elevator (high power for fast, repeatable height changes).	DRIVETRAIN Swerve.
ENDGAME Cage climb.	AUTONOMOUS Star-wheel outtake flips coral to L4 in auto; 6-motor elevator for fast height changes.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN CD CAD release

LESSONS FOR STUDENTS

- Star wheels can re-orient a game piece during ejection (flip coral up for L4).
- Sometimes a passive ramp beats a powered belt for straightening/holding a piece.

CD thread: <https://www.chiefdelphi.com/t/503355> · CD CAD release · Onshape CAD

★ Study pick

1156 Under Control (2025)

BRA

EPA pending

Swerve coral robot; permanent orientation-PID heading hold; tech binder

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	INTAKE Coral intake (see CAD/binder).
ELEVATOR / LIFT Elevator superstructure.	DRIVETRAIN Swerve that always runs with an orientation (heading) PID engaged.
ENDGAME Cage climb.	AUTONOMOUS Always-on heading PID makes auto-aim deterministic; coral auto (count unconfirmed).
VISION Unconfirmed (check binder)	PUBLIC CAD / DESIGN CD CAD, code & tech-binder release

LESSONS FOR STUDENTS

- Always-on heading PID makes driving and auto-aim more deterministic.
- A released technical binder is a strong reference for reproducibility.

CD thread: <https://www.chiefdelphi.com/t/505398> · CD CAD, code & tech binder · Onshape CAD

Coral + algae robot; 118 also author the 2025 Everybot

SCORING (CORAL / ALGAE) End-effector versatile enough to handle either CORAL or ALGAE so it fits any alliance role; full mechanism detail in the released STEP/IGES CAD.	INTAKE Coral / algae intake — the Everybot can pick up both game pieces.
ELEVATOR / LIFT Elevator / lift to the versatile end-effector (see CAD).	DRIVETRAIN Swerve.
ENDGAME Cage climb (the Everybot build includes a cage climber).	AUTONOMOUS Autonomous coral scoring; routine specifics not detailed.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN GrabCAD release

LESSONS FOR STUDENTS

- Robonauts publish their full competition-robot CAD (STEP/IGES) AND lead the Everybot program — a clone-friendly reference at two levels.
- Compare 'Firefly' against the accessible 2025 Everybot to see what a top program adds over the canonical low-cost build.

GrabCAD: <https://grabcad.com/library/robonauts-118-2025-robot-firefly-1> · CD reveal thread · [GrabCAD release](#)

Coral robot: ground coral intake -> elevator -> end effector, plus a climber; CAD + tech binder

SCORING (CORAL / ALGAE) Coral end-effector for reef placement (detail in CAD/binder).	INTAKE Coral GROUND intake (swapped to mid-season; hub secured with a set screw after testing).
ELEVATOR / LIFT Elevator / lift to the end effector (see CAD / tech binder).	DRIVETRAIN Swerve.
ENDGAME Cage climb — a climber fastened to the base plate (per the release thread).	AUTONOMOUS Autonomous coral scoring; routine specifics not detailed.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A full CAD + tech-binder release from a regional-championship (CHCMP) competitor is a reproducible reference for a competitive coral cycle.
- The thread documents practical reliability fixes (set-screwing the ground-intake hub).

Onshape CAD: <https://cad.onshape.com/documents/2bae4024a467119e47dc96b0/w/6c850cd2bf85d2162a5bb7d6/e/b4c78beff50beb69b98ad0b6> · CD CAD/code release · [Onshape CAD](#)

Coral + algae robot: full-width intake -> star-wheel indexer -> 2-stage cascade elevator + pivot -> dual coral/algae end effector, plus a deep-cage climber; tech binder

SCORING (CORAL / ALGAE) One Kraken X60 end effector that places CORAL and grabs ALGAE in quick succession (Ri3D-inspired); 3D-printed gear/pulley combos; 5.55:1 coral / 6.66:1 algae.	INTAKE Full-width intake; free-pivoting front roller picks coral from any angle; polycarb wedges center it; silicone-wrapped TPU rollers; dual #25-chain deploy (65.33:1). A carbon-fiber L1 bar (Kraken X44 + sector gear) scores L1. Star-wheel indexer orients coral and auto-ejects doubles via beam-break.
ELEVATOR / LIFT 2-stage cascade elevator (2x Kraken X60, 7:1, full height in 0.5 s; constant-force-spring counterbalanced) + a 52:1 pivot on a 3.5" x-contact bearing carrying the end effector.	DRIVETRAIN WCP X2i 'dropped swerve' (X60 drive / X44 steer, 15 ft/s); low-CG tube drop; bottom-facing electronics for serviceability; 29.5" frame.
ENDGAME Deep-cage climb: a 6 lb climber with 'Spear' + 'Fly Swatter' centering plates; torsion-spring deploy, a 25:1 Maxplanetary spools string to lift the robot in ~1.5 s (Ri3D-inspired).	AUTONOMOUS Autonomous coral scoring; routine specifics not detailed (the binder documents mechanisms).
VISION Cameras on the front swerve modules detect reef AprilTags; a 'Pi Motel' case houses 4x Rock 5C coprocessors with a custom power-over-PCIe PCB.	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- Bottom-facing electronics plus a 'dropped swerve' keep the robot serviceable and low-CG.
- A single-motor end effector that places coral AND grabs algae in quick succession is an efficient cycle (5.55:1 / 6.66:1 split).
- The 'Pi Motel' (4x Rock 5C coprocessors, power-over-PCIe PCB, cameras on the swerve modules) is a scalable, serviceable vision approach.

Onshape CAD: <https://cad.onshape.com/documents/96a5f9f437337066b2987626/w/9644a0c5069d0349ab3cf5a6/e/01df3e1fb18f1f03077c9c5d> · CD CAD/code release · Onshape CAD

Elevator-based coral robot that also REMOVES ALGAE; Open Alliance build blog; deep cage climb

SCORING (CORAL / ALGAE) Coral end-effector; the robot also removes ALGAE from the reef (needed for the coral ranking point).	INTAKE Coral intake; the robot also removes algae from the reef (for the coral ranking point).
ELEVATOR / LIFT Elevator superstructure (7407 reliably 'builds an elevator' every season).	DRIVETRAIN Swerve.
ENDGAME Deep cage climb — pursued for its high point value and the barge RP (inspired by Ri3d).	AUTONOMOUS Autonomous coral scoring; routine specifics not detailed.
VISION Unconfirmed (check CAD)	PUBLIC CAD / DESIGN Onshape CAD

LESSONS FOR STUDENTS

- A polished, well-presented CAD release + Open Alliance build blog is a model for documentation and communication.
- Follow the build blog to see how a ~30-student boarding-school team iterates across a season.

Onshape CAD: <https://cad.onshape.com/documents/c2395e8cc25ba2e37f225c1c/w/bd65b2a0c5d2dab99556c06b/e/9397772cbfdc7ca9f11dfdec> · CD reveal thread · Onshape CAD

Additional well-documented robots for study

Each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. ★ marks the recommended study picks.

Sources: the Spectrum 3847 CAD Collection and team CAD releases (Onshape, GrabCAD), Chief Delphi reveal / CAD-release threads, and match data from The Blue Alliance. EPA shown is Local EPA (approx), independently computed — not official Statbotics. "Unconfirmed" means a detail was not publicly published, not that it is absent.